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Neuroleptic Discontinuation in the Very Stable Schizophrenic Patient: Relapse Rates and Serum Neuroleptic Levels

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This double-blind, placebo-controlled study investigated the effect of discontinuation of maintenance neuroleptic medication in chronic, long-stay schizophrenic patients who had been clinically stable for over 5 years. After 12 months, 75 per cent of the patients on placebo had relapsed as compared to 33 per cent on active medication and relapses in the placebo group occurred significantly earlier during the study; survival analysis revealed the differences to be statistically significant. No clear predictors of relapse could be identified, although patients who relapsed from either group were more likely to have had higher scores for anxiety symptoms and a lower serum prolactin level at baseline. It was concluded that even in older patients who have been chronically ill with prominent negative symptoms and living in a relatively undemanding environment on stable doses of neuroleptics, the risk of relapse remains unacceptably high after abrupt withdrawal of maintenance neuroleptic therapy.

KEY WORDS—Chronic schizophrenia, neuroleptic withdrawal, relapse, serum neuroleptic levels.

INTRODUCTION

The value of maintenance neuroleptic therapy for prevention of relapse in schizophrenia has been well documented both in follow-up (May and Tuma, 1964; Prien and Cole, 1968; Hirsch *et al.*, 1973, Crawford and Forrest, 1974; Hogarty *et al.*, 1974; Curson *et al.*, 1985) and dose reduction studies (Johnson *et al.*, 1983; Kane *et al.*, 1983; Kane, 1987; Marder *et al.*, 1987). An alternative strategy for studying the effectiveness of maintenance neuroleptic therapy in schizophrenia has involved withdrawal of medication in remitted patients. During the 1960s several studies were conducted in long-term hospitalized patients using oral neuroleptic withdrawal (Olson and Peterson, 1962; Greenberg and Roth, 1966; Prien *et al.*, 1968; Gottschalk *et al.*, 1970) but the results were sometimes equivocal (Tobias and McDonald, 1974).

The abrupt withdrawal studies involving depot medication, however, have consistently shown an increased risk of relapse (Hogarty *et al.*, 1976; Johnson *et al.*, 1976; Dencker *et al.*, 1980, 1986; Levine *et al.*, 1980; Wistedt *et al.*, 1981) although most have been restricted to short periods of pre-trial stability (6 – 24 months) and to outpatient samples (Kane, 1987) reflecting practical difficulties of working in the community and a shift in target population (Kiloh *et al.*, 1988). Cheung (1981), however, reported 30 patients who had been in remission for between 3 and 5 years with a view to determining the feasibility of stopping medication, and found a relapse rate of 62 per cent. Similarly, some of the patients reported by Johnson (1979) had been stable for 5 years, but in both these studies the samples had been obtained from outpatients.

The main objective of the present study was to investigate, in a controlled manner, the effects of abrupt discontinuation of maintenance neuroleptic therapy in chronic inpatients who had been relapse-free for over 5 years in an unchanging and socially undemanding environment. A secondary objective of the study was to explore the relationship (if any) between the changes in clinical state and serum neuroleptic levels after abrupt drug withdrawal.

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METHOD

Patient sample

Six hundred inpatients with a diagnosis of schizophrenia were initially screened by inspection of the case notes for confirmation of diagnosis and stability of clinical state as indicated by absence of clinical deterioration and/or an increase of neuroleptic medication. A group of patients were identified who had been clinically stable and had not had a ward change during the previous 5 years. The inclusion criteria required that the patient had a clear diagnosis of chronic schizophrenia on Research Diagnostic Criteria (Spitzer *et al.*, 1975). Presence of coexisting physical illness excluded patients from the study, as did a diagnosis of mental handicap, epilepsy or brain damage. All the patients entering the study were required to have been on fluphenazine decanoate only, and if they had been prescribed other neuroleptics, this had been on rare occasions for acute behavioural problems and for never more than 24 hours in any one month; similar considerations were required for other psychotropics. The project was approved by the Salford Ethical Committee and informed written consent was obtained from each patient.

Design of study

The study design involved a double-blind, placebo-controlled discontinuation of maintenance fluphenazine decanoate after an initial observation period of 12 months during which patients were followed up at monthly intervals for systematic assessment on all the parameters used in the main study including the clinical rating scales and serum assays to ensure that their clinical picture remained stable under trial conditions. Throughout this phase, patients were maintained on their regular fluphenazine dose only and no p.r.n. medication was permitted. Care was taken to ensure that patients' daily routine remained relatively unchanged. At the end of this phase, 24 patients (from the initial sample of 30), who had remained clinically stable, were randomly allocated to either a placebo group who received injections without active medication or a control group who continued to receive their usual dose of fluphenazine decanoate. The two groups were followed up for a further period of 12 months, during which blood level estimation and clinical assessments were continued at monthly intervals.

For the purpose of the study a patient was considered to have relapsed if there was at least a 25 per cent increase in the total score on Brief Psychiatric Rating Scale (BPRS) (Overall and Gorham, 1962) and if the ward sister/charge nurse felt that there had been a significant worsening in ward behaviour as seen on Psychotic Inpatient Profile (PIP) (Lorr and Vestre, 1968) at any point during the study. Patients who relapsed were assessed in detail and restarted on their original dose of fluphenazine decanoate injections and prescribed other medication as required; in the case of patients who relapsed in the control group whilst on active medication, additional oral chlorpromazine was prescribed according to clinical need until such time as the symptom exacerbation was controlled. The treatment codes for relapsed patients were revealed only to the responsible consultants.

The study medication was provided by E. R. Squibb and Son, and each patient had his/her medication prepared and labelled in advance for the total duration of the study. The placebo injections consisted of sesame oil without the active drug in a volume identical to the original volume of the injection which the patient regularly received during initial phase of the study and was identical in physical appearance to the normal fluphenazine preparation. The researchers were blind to patients' treatments throughout the course of the study.

Clinical assessment

The diagnosis of schizophrenia was confirmed on Research Diagnostic Criteria (RDC). Baseline assessments were conducted at month 0 before the patient entered the study. Clinical assessments were carried out at monthly intervals using the following rating scales:

(1) Brief Psychiatric Rating Scale (BPRS). This rating scale has several items relating to acute and chronic schizophrenic illness but also includes items of general psychopathology. These can be grouped into subsets for the purpose of analysis such as the negative subset which includes items of emotional withdrawal, motor retardation and blunted affect and the positive subset with suspiciousness, hallucinatory behaviour and unusual thought content. The affective subset used in this study included anxiety, tension, depression and guilt items. The scale and its subsets have been extensively validated in clinical research (Overall and Gorham, 1988).
(2) Psychotic Inpatient Profile (PIP) (Lorr and

Vestre, 1968). This is an elaborate scale for rating behaviour of psychotic patients on the wards, and is administered by the researcher in conjunction with reports of the nursing staff and carers of the patients. It has built into it subsets for observed psychotic behaviour, including items such as hallucinatory and delusional behaviour, depression, excitation, etc. This scale was included to ensure that ward behaviour between assessments could be documented for consideration of the total clinical picture of patients in the study.

The inter-rater reliability of the researchers was ensured by periodic sessions during which the researchers rated interviews with schizophrenic patients on videotapes; the clinical ratings were conducted by two of the authors (G.S. and A.S.) and the serum assays were undertaken in the Hope Hospital laboratory by J.K.

Serum assays

Laboratory personnel responsible for measurements were blind to all treatments. Serum neuroleptic levels were measured by the radioreceptor assay using haloperidol as internal standard; the results expressed in terms of 'neuroleptic units'. Serum prolactin levels were measured by radioimmunoassay using a kit obtained from Chelsea Hospital for Women (Dovehouse Street, London). The standardization of the assays for serum neuroleptics and prolactin, and their sensitivity and reproducibility, have been reported in our previous communication (Krska *et al.*, 1986).

Statistical analysis

Relapse rates were analysed using survival analysis, and are presented as the number of patients remaining unrelapsed in each group. Univariate statistics using *t*-tests and one-way analysis of variance were used for analysing the differences between groups defined by medication and relapse status for parametric data and Mann–Whitney U and Kruskal–Wallis tests for variables that did not satisfy the normal distribution criteria such as the individual items of BPRS. Correlations were computed between change in psychopathology and absolute levels of neuroleptic and prolactin levels and their rate of decrease after neuroleptic withdrawal.

RESULTS

Characteristics of patient groups

The randomization procedure resulted in the two groups being approximately similar for demographic and clinical characteristics (Table 1). There were more females in the placebo group as compared to the control group. The two groups were identical with respect to age, duration of illness and number of years they had been in clinical remission. The mean dose of neuroleptic they were receiving before entry into the study was higher in the control group, as were the mean neuroleptic levels.

Table 1. Neuroleptic withdrawal in chronic schizophrenics: characteristics of patient sample

	Placebo group	Control group
Age (years)	57.6 ± 06.1	57.1 ± 05.0
Duration of illness	30.9 ± 08.2	35.2 ± 05.8
Hospitalization (years)	24.2 ± 08.3	25.6 ± 09.1
Pre-trial stability	6.2 ± 03.1	7.7 ± 02.6
Neuroleptic dose	33.3 ± 32.6	50.4 ± 29.6
Neuroleptic levels (RRA)	11.4 ± 7.3	14.2 ± 26.3
Prolactin levels	510 ± 572	630 ± 719
Affective syndrome	4.6 ± 0.7	4.6 ± 0.7
Positive score	8.0 ± 2.6	7.1 ± 1.4
Negative score	6.8 ± 1.1	7.6 ± 1.6
Total BPRS score	23.9 ± 3.9	25.9 ± 2.7
Total PIP score	26.1 ± 10.8	29.9 ± 11.0

Values are means ± standard deviations; significance values obtained after one-way analysis of variance

Relapse rates

Clinical relapses occurred in both groups during the 12-month follow-up, although the numbers were much higher in the placebo group. Thus, although only four out of 12 patients relapsed in the control group, nine of the 12 patients in the placebo group had relapsed during this period. Simple cross-tabulation showed that, even without taking the timing of relapse into account, the relapse rates in the two groups were significantly different (chi-square = 4·196; $p = 0{\cdot}041$).

Patients who relapsed in the control group tended to do so later during the observation period; all four relapsing after the ninth month of follow-up. By contrast, seven of the nine patients who

relapsed in the placebo group did so within the first 4 months. Survival analysis highlighted this difference; life tables for the two groups defined by medication status showed that the cumulative proportion surviving and the probability density were differently distributed. Comparison of survival experience of the two groups, using Lee–Desu statistic, confirmed that the two groups were markedly different (L-D statistic = 8·73; $p = 0{\cdot}0031$). When relapses occurred in the placebo group they were rapid in onset, with deterioration occurring over a period of a few days; in only three patients did the full relapse take more than a week. The patients in the control group generally tended to have a more insidious relapse with several days of prodromal symptoms such as anxiety, depression and disturbed behaviour on the PIP scale.

Serum neuroleptic levels

At baseline, serum neuroleptic levels were slightly higher in the control group, reflecting, perhaps, the higher fluphenazine dosage (see Figure 1). When groups defined by relapse status were compared, there appeared to be no difference in serum levels at baseline. At the time of relapse, serum levels were significantly lower in patients in the placebo group. Serum neuroleptic levels showed marked individual variation between patients in the control group; the mean values, however, showed reasonable stability over the 12-month period. By contrast, the levels showed a rapid initial fall in the patients in the placebo group, the rate of decrease slowing after the first 2 months, and by the ninth month no neuroleptic was detectable in the serum of any of these patients. The difference in serum neuroleptic levels between the two groups was statistically significant from the first month onwards. Although decreasing levels were associated with higher frequency of relapses in the placebo group, there was no direct correlation between the time of occurrence of relapse and the actual serum neuroleptic level. The rate of drop in serum levels was estimated by calculating the proportional change from baseline of serum neuroleptic levels for each month over the study period; the rate of decrease was highest in the first 3 months but showed no correlation with number of relapses or change in psychopathology. Thus, of the nine relapses in the placebo groups, two occurred after only 10 per cent reduction of serum neuroleptic level from baseline (at month 1 and 2), two after 50 per cent reduction (at month 3 and 4) and the remainder after over 80 per cent reduction (one at month 3, and the remainder after the fifth month); the period between relapse and the time when serum level had reached below 80 per cent of baseline level varied from 1 to 4 months. There was no change in the neuroleptic levels of patients in the control group who relapsed, indicating that factors other than neuroleptic action were operating in the precipitation of relapse.

Serum prolactin levels paralleled changes in serum neuroleptic levels during the study. In the control group the levels remained relatively stable throughout the 12 months of follow-up, whereas there was a drop in the prolactin levels in the patients in the placebo group corresponding to the decrease in neuroleptic levels. The difference in serum prolactin levels between the two groups achieved statistical significance at month 2, and remained so until the end of the study. When the initial serum prolactin levels were compared for groups defined by relapse status, it was found that patients who relapsed from either group had significantly lower prolactin levels as compared to those who did not; there was no difference in baseline prolactin levels between patients who relapsed on active medication as compared to those who did so on placebo. The timing of relapse did not correlate with serum prolactin levels, although patients who relapsed in the control group had significantly lower prolactin levels at the time of relapse than those who did not. In the placebo group two relapses occurred after only 10 per cent reduction of prolactin levels from baseline (at month 1 and 2), three after a 50 per cent reduction (at months 3, 5 and 8) and the remainder after 80 per cent reduction, indicating a lack of any direct relationship between timing of relapse and magnitude of fall in serum prolactin levels.

Characteristics of relapse

Depending on the relapse status, three groups were identified (Table 2): patients who relapsed after drug discontinuation, patients who relapsed on active medication and patients who did not relapse. There was no difference between the three groups for age, sex, duration of illness or hospitalization. Comparison of baseline clinical pictures of patients in the three groups revealed that, contrary to expectations, patients who did not relapse had a higher total PIP score, indicating more disturbed ward behaviour. There was no difference between the three groups for total BPRS scores or for positive

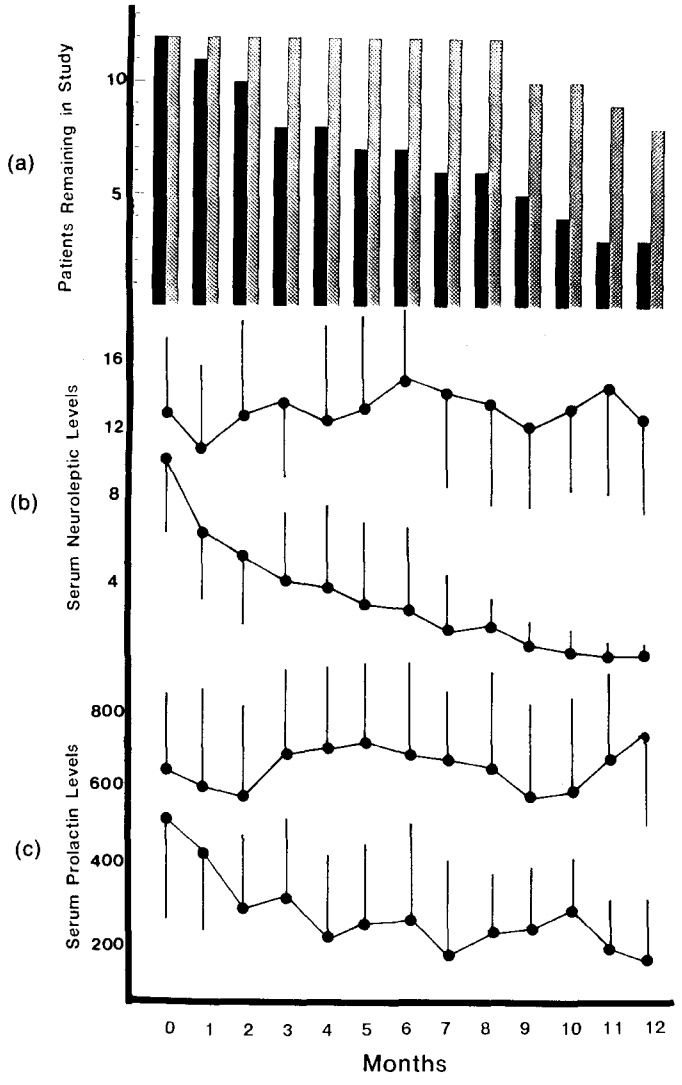


Figure 1. Depot neuroleptic withdrawal: (a) relapse rates (bars represent number of patients remaning unrelapsed in the study; (b) serum neuroleptic levels (neuroleptic units); (c) serum prolactin levels (IU/l). Values are means; vertical lines represent standard errors

or negative subsets. There were, however, significant differences between the groups with respect to affective symptoms; patients who eventually relapsed from either group had significantly higher scores on anxiety and tension items of BPRS.

The clinical change at the time of relapse suggested that in most patients there was a global deterioration of symptoms affecting all areas of psychopathology (Table 3), although individual patients showed considerable variability with respect to symptom patterns. There was no significant difference between the two relapse groups for clinical characteristics of relapse; severe deterioration of total BPRS and PIP scores was noticed in all relapsed patients from both groups, which in many cases more than doubled. The negative symptoms also showed deterioration, and in three patients the clinical change could be accounted for

Table 2. Neuroleptic withdrawal in chronic schizophrenics

		No relapse	Relapse (active)	Relapse (placebo)	Significance *p*
Age (years)	A	57.4 ± 5.4	57.5 ± 4.1	53.0 ± 6.7	ns
Duration of illness	A	32.0 ± 5.1	29.8 ± 6.7	31.6 ± 8.3	ns
Hospitalization (years)	A	24.1 ± 8.3	25.5 ± 4.8	25.7 ± 9.5	ns
Pre-trial stability	A	7.1 ± 3.1	8.2 ± 2.1	7.8 ± 2.9	ns
Neuroleptic dose (FPZ mg/2 weeks)	A	56.8 ± 39.7	56.3 ± 17.2	30.6 ± 27.3	ns
	B	56.8 ± 39.7	56.3 ± 17.2	0.0 ± 0.0	0.0029
Neuroleptic levels (RRA)	A	11.2 ± 27.4	10.4 ± 10.3	9.8 ± 3.7	ns
	B	12.4 ± 16.0	10.6 ± 9.8	1.9 ± 4.5	0.0021
Prolactin levels	A	967 ± 555	390 ± 185	331 ± 169	0.0491
	B	755 ± 766	431 ± 123	110 ± 66	0.0438

Values are means ± standard deviations; significance values obtained after one-way analysis of variance; A = baseline scores; B = scores at relapse

Table 3. Neuroleptic withdrawal in chronic schizophrenics: clinical characteristics at relapse (BPRS)

		No relapse	Relapse (active)	Relapse (placebo)	Significance *p*
Affective syndrome	A	4.3 ± 0.5	5.3 ± 0.5	4.7 ± 0.7	ns
	B	4.5 ± 0.7	10.8 ± 1.3	7.8 ± 2.2	0.0005
Positive score	A	7.7 ± 2.4	7.8 ± 1.0	7.2 ± 1.9	ns
	B	7.4 ± 2.3	14.5 ± 2.6	13.0 ± 3.8	0.0007
Negative score	A	7.7 ± 0.9	7.3 ± 2.6	6.4 ± 1.0	ns
	B	7.4 ± 1.2	10.3 ± 2.8	15.2 ± 4.6	0.0331
Total BPRS score	A	24.3 ± 3.9	25.5 ± 2.5	22.8 ± 2.7	ns
	B	23.8 ± 3.6	46.8 ± 7.9	42.7 ± 5.5	0.0001
Total PIP score	A	31.1 ± 11.5	19.5 ± 4.2	22.7 ± 9.3	ns
	B	28.0 ± 13.0	62.5 ± 7.5	63.2 ± 13.1	0.0001

Values are means ± standard deviations; significance values obtained after one-way analysis of variance; A = baseline scores; B = scores at relapse

by increase in mainly the anxiety and negative symptom scores, with little worsening of positive symptom psychopathology. On the PIP scale the relapsed patients showed a marked increase for items of disturbed behaviour such as irritability, uncooperativeness and aggression; several patients also showed withdrawal and dependency. When the two groups of relapsed patients were compared, affective symptoms appeared to be more prominent and negative symptom scores lower in patients who relapsed on active medication as compared to those who relapsed in the placebo group. Relapses in the placebo group tended to be more precipitate, whereas those in the control group were more insidious.

DISCUSSION

The design of the present study differed from previous depot neuroleptic withdrawal studies in several respects: first, unlike other studies to date, the patient sample selected consisted wholly of a very stable cohort whose illness had been in remission for many years before entering the study. It was therefore assumed that drug withdrawal may be justified in these patients and should be tested

under controlled conditions. Secondly, previous studies have not attempted to control for the numerous environmental factors known to be associated with clinical relapse (Schooler and Spohn, 1982), by using outpatients, short-term inpatients or patients in an uncontrolled milieu which may unpredictably affect the outcome. Thirdly, most of these studies have not included a period of assessment before instituting neuroleptic withdrawal to ensure clinical and behavioural stability; nor have they considered the constancy of neuroleptic medication before and during the period of observation. The present study was designed to take these factors into account.

It should be noted that the sample in our study relates to a very special group of patients who had been chronically ill and in remission for several years. It is therefore likely that the results from our study may not be universally applicable but, despite the difficulty in obtaining a sample of patients with prolonged clinical stability, this is the only group that can provide answers about the continuing need of neuroleptic medication in chronic stable schizophrenics. The decision to continue with maintenance neuroleptic medication in such patients is always difficult, and as yet no clear criteria have emerged on which to determine the safety of discontinuing the treatment, although several recommendations have been made (Johnson, 1979; Kissling, 1988). Earlier studies suggested that patients hospitalized for more than 15 years were less liable to relapse after discontinuation (Prien *et al.*, 1968). It was also suggested that patients in 'protected' environments and with prominent negative symptoms may not require maintenance medication (Leff, 1978). More recently it has been proposed that patients over the age of 40 have a better prognosis after withdrawal (Cheung, 1981). Our study, however, has shown that even in older patients who are chronically ill with prominent negative symptoms on an unchanging dose of neuroleptic and in prolonged remission, the risk of relapse remains unacceptably high when prophylactic medication is stopped. This implies that many patients with schizophrenia will probably need to be on prophylactic therapy for a long time; perhaps even for life.

One finding from the study that needs explanation is the occurrence of relapse amongst patients who were continued on active medication. This was surprising since these patients had been stable on the same dose of medication for several years before entering the study; the ward milieu and the hospital environment had remained relatively unchanged during this time. Previous withdrawal studies which have included control groups on active medication have also found that a proportion of patients (up to 25 per cent) relapse despite continuing treatment (Hogarty *et al.*, 1976; Wistedt, 1981). A similar figure is also found in follow-up studies of medicated schizophrenics in remission (Hirsch *et al.*, 1973; Hogarty, 1984; Curson *et al.*, 1985). One explanation for this observation may relate to the adverse effect of environmental influences in precipitating relapse. Thus, Wing *et al.*, (1964) found that increased therapeutic input in the form of industrial therapy for a group of chronic schizophrenics resulted in deterioration of symptoms and increased relapse rates. Similarly, in another study (Schooler and Spohn, 1982), increased interaction resulting from intensive socio-environmental therapy adversely affected the clinical state of chronic schizophrenic patients, leading to treatment failure and social dysfunction. Such an explanation can be valid in our study only if it is construed that the trial itself, by increasing contact, attention and interaction between patients and the researchers, had adversely affected their clinical state. Relapses in the control group occurred towards the end of the study, were more insidious in onset and had prominent affective symptoms, which suggests that the social interaction could have been important and perhaps cumulative over the months during which these patients had been intensively studied. It is possible that some of the relapses in the placebo group, specially those occurring late, may have had a similar pathogenetic contribution.

In common with some previous studies there was no direct relationship between serum neuroleptic levels and relapse; the frequency of relapse during any one month did not correlate with either absolute levels or with rate of decline in serum neuroleptic concentrations. We were unable to confirm earlier observations that patients who eventually relapse have lower serum dopaminergic binding whilst on stable doses of neuroleptic (Brown *et al.*, 1982), nor did our results show any relationship between relapse and change in neuroleptic levels after drug withdrawal (Wistedt *et al.*, 1982) except for a general observation that relapses were more likely to occur after serum levels had started to drop. It is possible that this may have been due to the small number of patients in the study.

Efforts at determining thresholds of serum neuroleptic concentrations below which relapses will

occur have produced conflicting findings (Wode-Helgodt *et al.*, 1978, Van Putten *et al.*, 1981; Harris *et al.*, 1982; Alfredsson *et al.*, 1984). Occasional reports of a therapeutic window (Casper *et al.*, 1980; Smith *et al.*, 1982; Garver *et al.*, 1984) with respect to neuroleptic treatment have not been consistently replicated. Recent reports using survival analysis with plasma fluphenazine as covariate suggested a high predictive value for schizophrenic relapse (Marder *et al.*, 1990). In our study some patients had a rapid reduction of serum neuroleptic concentration which stayed at very low levels for several months before relapse occurred. This suggests that although a relationship exists between low serum neuroleptic levels and propensity to relapse, the actual timing of the relapse cannot be accurately predicted from serum neuroleptic levels. Perhaps the main reason for the conflicting findings may relate to the nature of schizophrenic illness and its relapse, both of which have been poorly characterized, and the mechanisms by which neuroleptic exert their therapeutic and/or prophylactic control are not clearly understood.

Identification of factors which predict clinical relapse in schizophrenia could facilitate rationalization of therapy, and discontinuation could be undertaken with greater confidence and without undue risk to the patient. Attempts at such prognostication have so far been only partially successful (Prien *et al.*, 1968; Leff, 1978; Cheung, 1981). In the present study, patients who eventually relapsed, whether on active medication or on placebo, had higher baseline scores for anxiety and tension. It is possible that in these patients the psychotic anxiety may have been the only manifestation of a partially remitted illness which made them more vulnerable to relapse. The only other feature predictive of future relapse appeared to be a lower level of prolactin in patients from both groups who relapsed. Similar observations have been reported by Brown and Laughren (1981), but the significance of the finding and its clinical implications are at present unclear, since a subsequent study revealed exactly the opposite (Wilkins *et al.*, 1987).

In conclusion, our study has demonstrated the need for continuing medication for prevention of relapses, even for patients who have remained clinically stable for several years in a relatively undemanding environment. It follows that when patients' medication is reviewed with the view to discontinuation, the increased risk of relapse as a consequence must be taken into account before the decision is made.

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REFERENCES

Alfredsson, G., Bjkernstedt, L., Edman, G., Harnryd, C. and Oxistierna, G. (1984). Relationship between drug concentrations in serum and CSF, clinical effects and monoaminergic variables in schizophrenia treated with sulpiride and chlorpromazine. *Acta Psychiatrica Scandinavica* (Suppl. 311), **69**, 49–74.

Brown, W. A. and Laughren, T. (1981). Low serum prolactin and early relapse following neuroleptic withdrawal. *American Journal of Psychiatry*, **138**, (2), 237–239.

Brown, W. A., Laughren, T., Chisholm E. and Williams, B. W. (1982). Low serum neuroleptic levels predict relapse in schizophrenic patients. *Archives of General Psychiatry*, **39**, 998–1000.

Casper, R., Garver, D. L., Dekirmenjian, H., Chang, S. and Davis, J. M. (1980). Phenothiazine levels in plasma and red blood cells: their relationship to clinical improvement in schizophrenia. *Archives of General Psychiatry*, **37**, 301–305.

Cheung, H. K. (1981). Schizophrenics fully remitted on neuroleptic for 3–5 years—to stop or continue drugs? *British Journal of Psychiatry*, **138**, 490–494.

Crawford, R. and Forrest, A. (1974). Controlled trial of depot fluphenazine in outpatient schizophrenics. *British Journal of Psychiatry* **124**, 385–391.

Curson, D. A., Barnes, T. R., Bamber, R. W., Platt, S. R., Hirsch, S. R. and Duffy, J. C. (1985). Long-term depot maintenance of chronic schizophrenic outpatients: the seven year follow-up of the Medical Research Council fluphenazine/placebo trial I. Course of illness, stability of diagnosis and the role of a special maintenance clinic. *British Journal of Psychiatry*, **146**, 464–469.

Dencker, S. J., Malm, U. and Lepp, M. (1986). Schizophrenic relapse after drug withdrawal is predictable. *Acta Psychiatrica Scandinavica*, **73**, 181–185.

Dencker, S. J., Lepp, M. and Malm, U. (1980) Do schizophrenics well adapted in the community need neuro-

leptics? a depot neuroleptic withdrawal study. *Acta Psychiatrica Scandinavica*, Suppl. **279**, 64–76.

Davis, J. M. (1975). Overview: maintenance therapy in psychiatry: I. Schizophrenia. *American Journal of Psychiatry*, **132**, 1237–1245.

Garver, D. L., Hirschowitz, J., Glicksteen, G. A., Kanter, D. R. and Mavroidis, M. L. (1984). Haloperidol plasma and red blood cell levels and clinical antipsychotic response. *Journal of Clinical Psychopharmacology*, **4**, 133–137.

Gottschalk, L. A., Gleser, G. C., Gleghorn, J. M., Stone, W. N. and Winget, C. N. (1970). Prediction of changes in the severity of the schizophrenic syndrome with discontinuation and administration of phenothiazines in chronic schizophrenic patients: language as a predictor and measure of change in schizophrenia. *Comprehensive Psychiatry*, **11**, 123–140.

Greenberg, L. M. and Roth, S. (1966). Differential effects of abrupt versus gradual withdrawal of chlorpromazine in hospitalised chronic schizophrenic patients. *American Journal of Psychiatry*, **125**, 221–226.

Harris, P. Q., Friedman, M. J., Cohen, B. M. and Cooper, T. B. (1982). Fluphenazine blood levels and clinical response. *Biological Psychiatry*, **17** (10), 1123–1130.

Hirsch, S. R., Gaind, R., Rohde, P. D., Stevens, B. C. and Wing, J. K. (1973) Outpatient maintenance of chronic schizophrenic patients with long-acting fluphenazine: Double-blind placebo trial. *British Medical Journal*, **1**, 633–637.

Hogarty, G. E. (1984) Depot neuroleptics: the relevance of pychosocial factors—a United States perspective. *Journal of Clinical Psychiatry*, **45** (5), 36 – 42.

Hogarty, G. E., Goldberg, S. C. and the Collaborative Study Group (1974). Drug and sociotherapy in the aftercare of schizophrenic patients: one year relapse rates. *Archives of General Psychiatry*, **28**, 54–64.

Hogarty, G. E., Ulrich, R. F., Mussare, F. and Aristigueta, N. (1976). Drug discontinuation among long-term successfully maintained schizophrenic outpatients. *Diseases of the Nervous System*, **37**, 494–500.

Johnson, D. A. W. (1979). Further observation on the duration of depot neuroleptic maintenance therapy in schizophrenia. *British Journal of Psychiatry*, **135**, 524–530.

Johnson, D. A. W., Pasterski, G., Ludlow, J. M., Street, K. and Taylor, R. E. W. (1983). The discontinuance of maintenance neuroleptic therapy in chronic schizophrenic patients: drug and social consequences. *Acta Psychiatrica Scandinavica*, **67**, 339–352.

Kane, J. (1987). Treatment of schizophrenia. *Schizophrenia Bulletin*, **13** (1), 133–156.

Kane, J. M., Rifkin, A., Woerner, M. and Sarantakos, S. (1983). Low dose neuroleptic treatment of outpatient schizophrenics: 1. Preliminary results of relapse rates. *Archives of General Psychiatry*, **40**, 893–896.

Kane, J. M., Rifkin, A., Woerner, M. Reardon, C., Kreisman, D., Blumenthal, R. and Borenstein, M. (1986). High-dose versus low-dose strategies in the treatment of schizophrenia. *Psychopharmacology Bulletin*, **21**, 533–537.

Kiloh, L. G., Smith, J. S. and Johnson, G. F. (1988). *Physical Treatments in Psychiatry*. Blackwell Scientific Publications: Melbourne.

Kissling, W. (1988). Consensus regarding indications for prophylactic neuroleptic treatment—necessary but unattainable. In: *Depot Neuroleptics: A Consensus*, Barnes, T. R. E. (ed.). Mediscript, London, pp. 41–46.

Krska, J., Sampath, G., Shah, A. and Soni, S. D. (1986). Radioreceptor assay of serum neuroleptic levels in psychiatric patients. *British Journal of Psychiatry*, **148**, 187–193.

Leff, J. (1978). Psychophysiological monitoring of social and drug effects in schizophrenia. In: *Depot Fluphenazines: 12 Years of Experience*, Ayd, F. (ed.). Ayd Medical Communications, pp. 33–46.

Levine, J., Schooler, N. R., Severe, J., Escobar, J., Mandel, M., Sovner, R. and Steinbrook, R. (1980). Discontinuation of oral and depot fluphenazine in schizophrenic patients after one year of continuous medication: a controlled study. In: *Long-Term Effects of Neuroleptics (Adv. Biochem Psychopharmacol.*, Vol. 24), Cattabeni, F. (ed.). Raven Press, New York, pp. 483–495.

Lorr, M. and Vestre, N. D. (1968). *Psychotic Inpatient Profile*. Western Psychological Services, Los Angeles, CA.

Marder, S. R., Van Putten, T., Mintz, J., Lebelle, M., McKenzie, J. and May, P. R. A. (1987). Low dose and conventional dose maintenance therapy with fluphenazine decanoate: two year outcome. *Archives of General Psychiatry*, **44**, 518–521.

Marder, S. R., Van Putten, T., Aravagiri, M., Hawes, E. M., Hubbard, J. W., McKay, G., Mintz, J. and Midha, K. K. (1990). Fluphenazine plasma levels and clinical response. *Psychopharmacology Bulletin*, **26** (2), 256–259.

May, P. R. A. and Tuma, A. H. (1964). The effect of psychotherapy and stelazine on length of hospital stay, release rate and supplemental treatment of schizophrenic patients. *Journal of Nervous and Mental Diseases*, **139**, 362–369.

Olson, G. W. and Peterson, D. B. (1960). Sudden removal of tranquilising drugs from chronic psychiatric patients. *Journal of Nervous and Mental Disease*, **131**, 252–259.

Overall, J. E. and Gorham, D. R. (1962). The brief psychiatric rating scale. *Psychological Reports*, **10**, 799–812.

Overall, J. E. and Gorham, D. R. (1988). The Brief Psychiatric Rating Scale (BPRS): recent developments in ascertainment and scaling. *Psychopharmacology Bulletin*, **24** (1), 97–124.

Prien, R, and Cole, J. (1968). High dose chlorpromazine therapy in chronic schizophrenia. Report of National

Institute of Mental Health Psychopharmacology Research Branch Collaborative Study. *Archives of General Psychiatry*, **18**, 482–495.

Prien, R., Cole, J. O. and Belkin, N. (1968). Relapse in chronic schizophrenics following abrupt withdrawal of tranquilising medication. *British Journal of Psychiatry*, **115**, 679–686.

Renton, C. A., Affleck, J. W. and Carstairs, G. M. (1963) A follow-up of schizophrenic patients in Edinburgh. *Acta Psychiatrica Scandinavica*, **39**, 548–600.

Schooler, C. and Spohn, H. E. (1982). Social dysfunction and treatment failure in schizophrenia. *Schizophrenia Bulletin*, **8** (1), 85–98.

Smith, R. C., Vroulis, G., Shvartsburd, A., Allen, R., Lewis, N., Schooler, J. C., Chojnacki, M. and Johnson, R. (1982). RBC and plasma levels of haloperidol and clinical response in schizophrenia. *American Journal of Psychiatry*, **139** (8), 1154–1156.

Spitzer, R. R., Endicott, J. and Robins, E. (1975). *Research Diagnostic Criteria. Instrument No. 58*. New York State Psychiatric Institute, New York.

Tobias, L. L. and MacDonald, M. L. (1974). Withdrawal of maintenance drugs in long-term hospitalised mental patients: a critical review. *Psychological Bulletin*, **81** (2), 107–125.

Van Putten, T., May, P. R. A. and Jenden, D. J. (1981). Does a plasma level of chlorpromazine help? *Psychological Medicine*, **11**, 729–734.

Wilkins, J. N., Marder, S. R., van Putten, T., Midha, K. K., Mintz, J., Setoda, B. S., May, P. (1987). Circulating prolactin predicts risk of exacerabation in patients on depot fluphyenazine. *Psychopharmacology Bulletin*, **23**, 522–525.

Wing, J. K., Bennett, D. H. and Denham, J. (1964). The industrial rehabilitation of long-stay schizophrenic patients. *Medical Research Council Memo No. 42*. HMSO, London.

Wistedt, B. (1981). A depot withdrawal study. A controlled study of the clinical effects of withdrawal of depot fluphenazine decanoate and flupenthixol decanoate in chronic schizophrenic patients. *Acta Psychiatrica Scandinavica*, **65**, 65–84.

Wistedt, B., Jorgensen, A. and Wiles, D. A. (1982). A depot neuroleptic withdrawal study: plasma concentrations of fluphenazine and flupenthixol and relapse frequency. *Psychopharmacology*, **78**, 301–304.

Wode-Helgodt, B., Borg, S., Fyro, B. and Sedvall, G. (1978). Clinical effect and drug concentrations in plasma and cerebrospinal fluid in psychotic patients treated with fixed doses of chlorpromazine. *Acta Psychiatrica Scandinavica*, **58**, 149–173.